

RNA Extraction from Tasso Blood Samples: In-Flight Protocol

Name: _____ CWID: _____

Overview

In this lab, you will extract RNA from a simulated in-flight spaceflight blood sample collected using the Tasso device. You'll follow three main steps:

1. Erythrocyte lysis using the NEB Monarch RBC Lysis Buffer protocol
2. RNA isolation using the Qiagen RNeasy Mini Kit
3. RNA quantification using the Qubit 4 Fluorometer

You'll also learn about Tapestation and Fragment Analyzer quality control methods.

Part 1: RBC Lysis Protocol (Monarch)

Q: Why is it important to remove red blood cells before RNA extraction from whole blood?

A: _____

Materials

- Tasso blood sample (500 µl)
- RBC Lysis Buffer (cold)
- PBS (cold)
- 2 ml microcentrifuge tubes

Steps

1. Add 1.5 ml cold RBC Lysis Buffer to 500 µl whole blood in a 2 ml tube.
 2. Close the tube and invert 3–5 times to mix.
 3. Incubate on ice for 6–12 minutes. Mix occasionally. Sample should turn translucent.
 4. Centrifuge at $1000 \times g$ for 3 min at 4°C.
 5. Remove supernatant, leaving ~15–20 µl behind. Avoid disturbing the pellet.
 6. Flick the tube or vortex briefly to resuspend the pellet.
 7. Add 1.5 ml cold RBC Lysis Buffer, vortex to resuspend, and incubate on ice for 3 min.
 8. Repeat centrifugation and remove supernatant as before.
 9. Add 1.5 ml cold PBS, vortex to mix.
 10. Centrifuge again at $1000 \times g$ for 3 min. Remove supernatant, leaving 15–20 µl.
- SAVE THE PELLETT: You will proceed directly to RNA extraction.

Part 2: RNA Isolation (Qiagen RNeasy Mini Kit)

Goal: Extract total RNA from white blood cells using silica column-based purification.

Q: What is the function of each of the following components:

- **Buffer RLT:** _____
- **70% Ethanol:** _____
- **Buffer RW1:** _____
- **Buffer RPE:** _____
- **RNase-free water:** _____

Materials

- Qiagen Buffer RLT (with β -ME)
- 70% Ethanol
- Buffer RW1, RPE
- RNase-free water
- RNeasy spin column and collection tubes

Steps

1. Add 350 μ l Buffer RLT + β -ME to the pellet and vortex to lyse the cells.
2. Add 350 μ l of 70% ethanol. Mix well by pipetting.
3. Transfer the mixture to a spin column in a collection tube.
4. Centrifuge at $\geq 8000 \times g$ for 15 s. Discard flow-through.
5. Add 700 μ l Buffer RW1, centrifuge (15 s). Discard flow-through.
6. Add 500 μ l Buffer RPE, centrifuge (15 s). Discard flow-through.
7. Add 500 μ l Buffer RPE, centrifuge 2 min to dry the column.
8. Optional: Spin again in new tube to remove residual ethanol.
9. Place column in a clean 1.5 ml tube. Add 30–50 μ l RNase-free water to the membrane.
10. Centrifuge 1 min to elute the RNA. Store on ice.

Part 3: RNA Quantification (Qubit RNA HS Assay)

Goal: Use Qubit to determine RNA concentration via fluorescence (more accurate than Nanodrop at low concentrations).

Q: What is the purpose of including standards in Qubit quantification?

A: _____

Materials

- Qubit RNA HS Kit (reagent, buffer, standards)
- 0.5 ml Qubit assay tubes
- Qubit 4 Fluorometer

Steps

1. Label two tubes for standards (Standard 1 and 2) and one for each RNA sample.
2. Prepare Qubit working solution: Mix 199 μ l buffer + 1 μ l reagent per tube. Make enough for standards + samples.
3. Add: 190 μ l working solution + 10 μ l standard to each standard tube; 180–199 μ l working solution + 1–20 μ l sample to each sample tube.
4. Vortex all tubes for 3–5 seconds.
5. Incubate 2 min at room temperature.
6. Insert tubes into the Qubit and record RNA concentrations.

Part 4: Advanced RNA Quality Control

Tapestation (Agilent)

Purpose: Determine RNA integrity via electrophoretic separation. Produces an RNA Integrity Number (RIN).

- RIN 9–10 = Excellent RNA (intact 28S and 18S rRNA peaks)
- Degraded RNA appears as smears or with loss of rRNA peaks
- Can quantify RNA and visualize fragment distribution

Why use it? It's fast, uses low input, and gives both quantity and quality.

Fragment Analyzer (Agilent / Advanced Analytical)

Purpose: Similar to Tapestation but more high-throughput and automated. Also generates RNA Quality Number (RQN) or DV200 score.

- DV200 = % RNA fragments > 200 nucleotides
- Excellent for checking RNA for sequencing or library prep compatibility

Why use it? It can analyze many samples at once with high precision and greater resolution.

Post-Lab Questions

Q: How might microgravity affect standard phlebotomy methods, and why are devices like Tasso used instead?

A: _____

Q: Why do we use β -mercaptoethanol (or DTT) in the lysis buffer, and what might happen if you forget it?

A: _____

Q: If you accidentally disturbed the pellet during lysis, what downstream effect would you expect?

A: _____

Q: What would happen if you used too much or too little ethanol during the RNA binding step?

A: _____

Q: What might cause a poor RNA yield even if all visible steps were followed correctly?

A: _____

Q: How does Qubit differentiate between RNA and other nucleic acids like DNA?

A: _____

Q: You obtain a Qubit concentration of 12 ng/ μ l. If you eluted in 40 μ l, what is your total RNA yield?

A: _____